

Physical quantity	Component	Direction
Scalar	1 Component	No direction (no unit vector)
Vector	Each Component	1 direction (one unit vector)
Tensor	Each Component	More than one direction (more than one unit vectors)

Physical quantity	Component	Rank
Scalar	1 Component with zero direction	Zero
Vector	Each Component has one direction	One
Tensor of rank two	Each Component associated with two directions	Two
Tensor of rank three	Each Component associated with three directions	Three
Tensor of rank n	Each Component associated with n directions	n

(c) Intensity of magnetisation

Any bulk material (any object of finite size) contains a large number of atoms. Each atom consists of electrons which undergo orbital motion. Due to orbital motion, electron has magnetic moment which is a vector quantity. In general, these magnetic moments orient randomly, therefore, the net magnetic moment is zero per unit volume of the material.

When such a material is kept in an external magnetic field, atomic dipoles are created and hence, it will try to align partially or fully along the direction of external field. The net magnetic moment per unit volume of the material is known **as intensity of magnetisation or magnetisation vector or magnetisation.** It is a vector quantity. Mathematically,

$$\vec{M} = \frac{magnetic\,moment}{volume} = \frac{1}{V}\vec{p}_m \qquad (3.33)$$

The SI unit of intensity of magnetisation is ampere metre^{-1}. For a bar magnet of pole strength q_m, length $2l$ and area of cross-section A, the magnetic moment of the bar magnet is $\vec{p}_m = q_m \overrightarrow{2l}$ and volume of the bar magnet is $V = A\left|\overrightarrow{2l}\right| = 2lA$. The intensity of magnetisation for a bar magnet is

$$\vec{M} = \frac{magnetic\,moment}{volume} = \frac{q_m \overrightarrow{2l}}{2lA} \qquad (3.34)$$

In magnitude, equation (3.34) is

$$\left|\vec{M}\right| = M = \frac{q_m \times 2l}{2l \times A} \Rightarrow M = \frac{q_m}{A}$$

This means, **for a bar magnet the intensity of magnetisation can be defined as the pole strength per unit area (face area).**

(d) Magnetic induction or total magnetic field

When a substance like soft iron bar is placed in an uniform magnetising field $\vec{H}$, it becomes a magnet, which means that the substance gets magnetised. The magnetic induction (**total magnetic field**) inside the specimen $\vec{B}$ is equal to the sum of the magnetic field $\vec{B}_o$ produced in vacuum due to the magnetising field and the magnetic field $\vec{B}_m$ due to the induced magnetisation of the substance.

$$\vec{B} = \vec{B}_o + \vec{B}_m = \mu_o \vec{H} + \mu_o \vec{I}$$

$$\Rightarrow \vec{B} = \vec{B}_o + \vec{B}_m = \mu_o(\vec{H} + \vec{I}) \qquad (3.35)$$

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